

16-Tap FIR Low Pass Filter

RTL-GDSII

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1 Objective

Design, simulate, and analyze the behavior of a 16 tap FIR Low Pass Filter as an application example of the complete RTL-to-GDSII ASIC design flow using Synopsys EDA tools.

NOTE: The filter is intentionally limited to sixteen taps in order to reduce hardware complexity while still demonstrating the complete digital implementation.

2 Specifications

Parameter	Specification
Filter type	Low-pass FIR
Number of taps	16
Design method	Hamming window
Sampling frequency (f_s)	100 kHz
Nyquist frequency (f_N)	50 kHz
Cutoff frequency (f_c)	12.5 kHz
Normalized cutoff frequency	$0.25 f_N$ ($0.125 f_s$)
Input word length	16-bit signed fixed-point
Coefficient word length	16-bit signed fixed-point
Technology	GlobalFoundries 180 nm Standard Cell Library
Implementation flow	RTL $\rightarrow$ GDSII ASIC Flow

Table 1: Design specifications of the proposed FIR filter.

3 Design and implementation

3.0.1 Theoretical filter design

A Filter is a simple device that allows altering the frequency content of a signal, and can be represented as mathematical expression with certain properties. For digital signals the equivalent model is established by using discrete time functions.

Properties of filters are better understood as functions of frequency domain. This functions doesn't introduce new frequencies meaning that they are linear time-invariant systems.

By function, they are classified as high pass, low pass, band stop, or band pass. In this case, the focus remains in the low pass filter which purpose is keep frequencies below certain value, called cutoff frequency , and attenuated or suppress the values above it.

To obtain a good model function for such task and still be a realizable design, the function must be causal or, mathematically, satisfy the Paley–Wiener Theorem. This implies that the ideal filter is not realizable because the resulting time-domain function, sample function, is not causal. However, by truncating and applying a delay the function could be shifted and, thus satisfy this criteria.

Ideal filter function $H_d(\omega)$ has a instantaneous transition at ω_c resulting in a infinite time response with a decaying factor or a tail that never fully decays to zero. When a real filter is implemented, $h_d(n)$ is truncated resulting in some values of the decaying tail left off. This introduces a ringing close to ω_c known as Gibbs phenomenon. Note that no matter how many taps M are taken, the overshoot stays at a fixed 9% value of the jump height. Since truncation

is therefore unavoidable, the real filter function must have a finite transition (transition-band) between the desired frequencies (pass-band) and the attenuated frequencies (stop-band).

Filters can be also classified in two groups according to the response they produce in time-domain: Finite impulse response (FIR) and Infinite impulse response (IIR).

An IIR filter is defined mathematically as a discrete time function

$$y(n) = \sum_{k=0}^M b_n x(n-k) + \sum_{k=1}^N a_n y(n-k)$$

By nature the filter is recursive because of the term $\sum_{k=1}^N a_n y(n-k)$ implying the existence of a feedback path that could lead to unstable behavior due the presence of poles in Z-transform transfer function. However, they are widely used because of the reduction in needed coefficients to obtain a narrower transition-band that represent less memory usage and faster responses.

An FIR filter is represented as

$$y(n) = \sum_{k=0}^{M-1} b_k x(n-k) \quad (1)$$

Notice that there is no recursive term which lead to a always stable function or alternatively, no poles in Z-transform transfer function. This means that the filter only depends on current and past inputs.

Saying this, the analysis of the chosen topology, FIR Low-pass (LP) filter begins by looking the response of feeding the FIR equation with a impulse of $x(n) = \delta(n)$ results in

$$y(n) = \sum_{k=0}^{M-1} b_k \delta(n-k) = b_n$$

$\delta(n-k)$ is zero except when $n=k$, the impulse response are the coefficients itself. This is the unit sample response $h(n)$, so, by notation (1) becomes

$$y(n) = \sum_{k=0}^{M-1} h(k) x(n-k) \quad (2)$$

So the problem is reduced to determine the coefficients $h(n)$ for a desired frequency response $H_d(\omega)$. Before moving, another condition must be satisfied to provide correct characterization for a LP filter: if the symmetry condition

$$h(n) = h(M-1-n)$$

is satisfied the filter has a linear-phase or a constant group delay, ensuring the filter doesn't distort the signal.

Still one detail remain, and its that the resultant $h_d(n)$, from applying inverse DTFT to $H_d(\omega)$, is infinite in duration and must be truncated to a $n = M-1$, that is archive though Window Method, resulting in

$$h(n) = h_d(n)w(n)$$

$h(n)$ is a multiplication in time. In frequency this domain yields the convolution of $H_d(\omega)$ and $W(\omega)$, which function's spectrum has a main lobe that produces a transition band. Also this spectrum has sides lobes that produces ripples in pass and stop band.

The following $H_d(\omega)$ is proposed*,

$$H_d(\omega) = \begin{cases} e^{-j\omega \frac{M-1}{2}}, & 0 \leq |\omega| \leq \omega_c, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

which magnitude response follows the ideal rectangular window(keeping the amplitude of the signal in the pass-band) and, which frequency response presents a $\frac{M-1}{2}$ delay making it a causal function.

Applying inverse DTFT to (3)

$$h_d(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega(n - \frac{M-1}{2})} d\omega = \frac{\sin[\omega_c(n - \frac{M-1}{2})]}{\pi(n - \frac{M-1}{2})}, \quad n \neq \frac{M-1}{2}.$$

where ω_c is the normalized cutoff frequency that satisfies $0 < \omega_c < \pi$.

Notice that, for even M numbers the function never indeterminate because n are integer values.

By selecting a Hamming window, which has a wide main-lobe and low side-lobes,

$$w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{M-1}\right), \quad 0 \leq n \leq M-1.$$

the ideal magnitude response of $H_d(\omega)$ acquire a wide transition-band and small ripples in pass and stop bands.

With a even number of tap and the symmetric condition, at Nyquist frequency $\omega = \pi$,

$$H(\omega) = \sum_{n=0}^{M-1} h(n)e^{-j\omega n} \quad (4)$$

$$H(\pi) = \sum_{n=0}^{M-1} h(n)e^{-jn\pi} = \sum_{n=0}^{M-1} h(n)(-1)^n$$

Taking as a example M=4,

$$H(\pi) = h(0) - h(1) + h(2) - h(3)$$

Applying the symmetry condition yields $h(0) = h(3)$ and $h(1) = h(2)$, so

$$H(\pi) = 0$$

Its easy to see that coefficients cancels for all M evens because the interspersed signs and the symmetry of them, assuring that at the highest useful frequency, the filter suppresses the signal.

With both expressions established, to avoid tedious manual calculations, the computation of the filter coefficients was performed in Python.

3.1 Python Implementation

Note: The complete Python implementation is available in the project repository at <https://github.com/jebenitz/FIR-LP-Filter/blob/main/python/coefficients.py>

Due to the iterative nature of the coefficients calculations, the **numpy** library was used among **signal** (**scipy**) to obtain back $H(\omega)$, and **matplotlib** to plot the corresponding magnitude response $|H(\omega)|$.

A LP filter characteristic is that $H(0) \neq 0$. Evaluating equation (4) at zero yields,

$$H(0) = \sum_{n=0}^{M-1} h(n) = h(0) + h(1) + \dots + h(M-1)$$

For the case, the given 16 taps addition converges to a value close to 1. However, to make sure that exactly $H(0) = 1$ condition is met, the h expression was normalized to adjust the coefficients,

```
1 h = h / np.sum(h)    # normalize so H(0) = 1
```

Setting the parameters **fs**, **fc** and **M** to the corresponding values, the coefficients obtained by running the Python script (and saved in a .csv file) are presented in table 2.

Coefficient	Floating-point	Q1.15	Hexadecimal
$h(0)$	-0.001296	-42	0xFFD6
$h(1)$	-0.005406	-177	0xFF4F
$h(2)$	-0.012387	-406	0xFE6A
$h(3)$	-0.010745	-352	0xFEA0
$h(4)$	0.020421	669	0x029D
$h(5)$	0.090370	2961	0x0B91
$h(6)$	0.178421	5846	0x16D6
$h(7)$	0.240624	7885	0x1ECD
$h(8)$	0.240624	7885	0x1ECD
$h(9)$	0.178421	5846	0x16D6
$h(10)$	0.090370	2961	0x0B91
$h(11)$	0.020421	669	0x029D
$h(12)$	-0.010745	-352	0xFEA0
$h(13)$	-0.012387	-406	0xFE6A
$h(14)$	-0.005406	-177	0xFF4F
$h(15)$	-0.001296	-42	0xFFD6

Table 2: FIR filter coefficients in floating-point, Q1.15, and hexadecimal formats.

Before operating with this numbers at RTL level, the default floating-point is converted to a signed Q1.15 (fixed-point) format that allows a representation of values between -1.0 and +0.9999695, suitable for coefficients which tend to be in this range, with a precision of 2^{-15} in a finite space. To convert the values, the following expression was implemented

$$h_{fixed} = round(h \times 2^{15})$$

To make sure the specified 16-bit word-length doesn't wrap or overflow, the coefficients are first transformed to a 32-bit integer format and verified.

```
1 assert np.all(hq32 >= -32768) and np.all(hq32 <=
   32767)
```

If the condition is met, then the integer format is narrowed again to fit 16-bits, and ready to be converted to hexadecimal format

```
1 hhex = [format(int(v) & 0xFFFF, '04X') for v in hq15]
```

that is later used to generate a local parameter .vh file at RTL level.

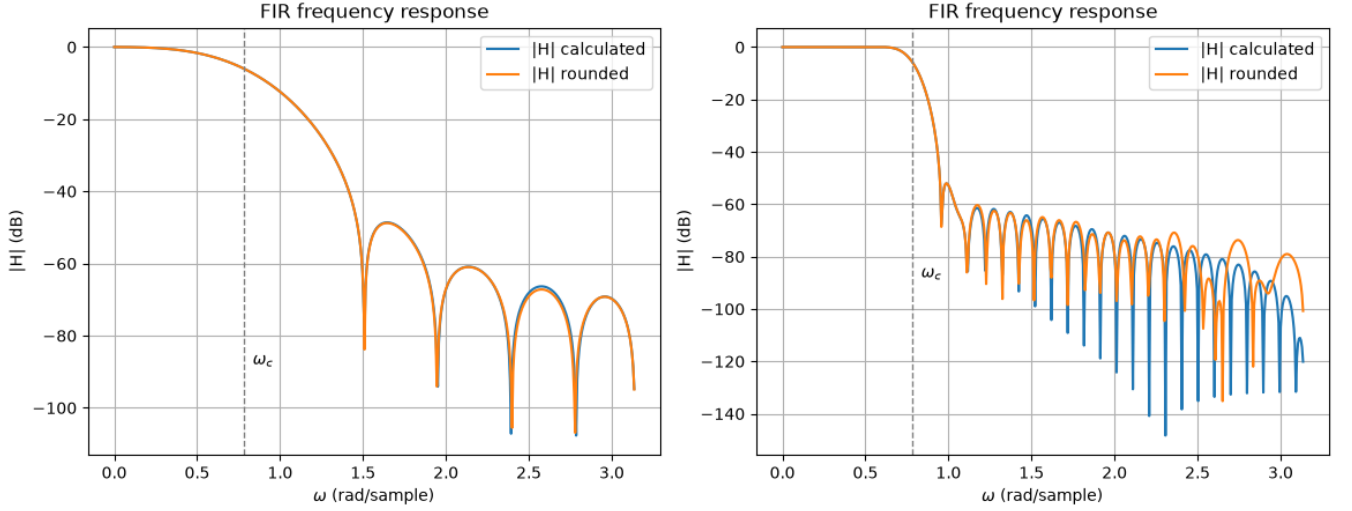
However, this Q1.15-formatted version of the coefficients carries an quantization error due to the preceding rounding step. To verify that this error doesn't modify in a significant way the magnitude response, the calculated and rounded responses are obtained

```
1 freqs, H = signal.freqz(h, worN=512)
2 freqs, HR = signal.freqz(hround, worN=512)
```

and plotted again each other, resulting in Figure 1a which confirms that 16-taps are sufficient to represent a LP filter. The two curves are almost visually indistinguishable and converge to similar values specially in pass and transition band.

However, as an illustrative example, the number of taps was changed to $M=64$ resulting in the plot shown in figure 1b.

Figure 1



(a) Magnitude response $|H(\omega)|$ comparison between the calculated and rounded coefficients with 16 taps

(b) Magnitude response $|H(\omega)|$ comparison between the calculated and rounded coefficients with 64 taps

In accordance with what was previously mentioned, as the number of taps is increased, the transition-band becomes sharper. While in the stop-band the side-lobes narrow and become numerous, and the maximum values stay in the same range for $M=16$ and $M=64$.

Notice that effects of rounded coefficients in the 64-taps plot are now visible and significantly diverges from the calculated values in the stop-band. Consider that as the number of taps is increased, the coefficients "redistribute" across more terms resulting in smaller values where the rounding step 2^{-15} becomes a significant error.